

Linux Command Line Mastery

Why Linux Is Non-Negotiable

Most servers, most cloud, and 99% of security tools run on Linux. If you can't operate a terminal, you can't be a cybersecurity professional.

File Navigation

```
pwd                # where am I
ls -la             # list everything with details
cd /var/log        # change directory
find / -name "*.conf" # search the file system
locate shadows     # locate files by name
file suspicious.bin # what kind of file is this?
```

File Permissions: The 10 Characters

`-rwxr-xr--` decodes as:

Type (`-` file, `d` dir, `l` symlink)

Owner: rwx

Group: r-x

Others: r--

```
chmod 750 script.sh # owner rwx, group r-x, others none
chown root:admin key # change owner:group
```

Security insight: world-writable files (`chmod 666`) and SUID binaries are classic privilege escalation targets. `find / -perm -4000` (setuid) reveals them.

Processes

```
ps aux                # all processes
top / htop            # live view
kill -9 <pid>         # force kill
pgrep -af "miner"     # find process by name
```

Dark knowledge: malware often names itself innocuously (e.g., `httpd`, `svchost`). You verify by `/proc/<pid>/exe` – the actual binary path.

Pipes, Redirection, and Text Tools

```
# Find anomalies in a log
grep -i "error" /var/log/syslog | sort | uniq -c | sort -rn
# Extract IPs from access log
grep "GET" access.log | awk '{print $1}' | sort -u
# Count login failures
cat auth.log | grep "Failed password" | wc -l
```

The "awk/sed/grep" trio is the engine of log analysis.

Persistence (Attackers Do This Too)

```
# list what runs at boot
ls -la /etc/init.d /etc/systemd/system
systemctl list-unit-files | grep enabled
cat /etc/crontab && ls -la /etc/cron*
```

Persistence = anything in these locations is executed on boot.
Defenders audit them; attackers plant in them.

[← Module start](#)

[Bash Scripting & Automation →](#)